

FACTORY REALLOCATION AND SHIPPING OPTIMIZATION RECOMMENDATION SYSTEM

*A Geospatial Decision Intelligence and Machine Learning Framework for
Confectionery Supply Chain Optimization*

UNIFIED MENTOR MACHINE LEARNING INTERNSHIP PROJECT REPORT

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DECLARATION OF AUTHENTICITY

I hereby declare that this project report titled “**Factory Reallocation and Shipping Optimization Recommendation System for Nassau Candy Distributor**” submitted to **Unified Mentor** represents my original work carried out as part of the Machine Learning Internship Program.

All algorithms, data engineering pipelines, mathematical modeling, simulation engines, and Streamlit application interfaces documented in this report were developed in accordance with the project guidelines. Appropriate citations and disclaimers have been provided for all external methodologies and reference materials.

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ABSTRACT

In modern wholesale distribution, static manufacturing allocation policies create substantial inefficiencies when customer demand shifts geographically. Nassau Candy Distributor operates five manufacturing and warehousing hubs across the United States. Historically, product lines were assigned to factories using rigid legacy rules, forcing high-volume confectionery lines to travel thousands of miles across the country.

This study implements an end-to-end Decision Intelligence and Optimization Recommendation System. Using 10,194 historical order transactions, we engineer geodesic Haversine transit distances and temporal lead times. We apply unsupervised K-Means clustering to uncover congested shipping corridors, train and evaluate multiple supervised machine learning regression models (Linear Regression, Ridge, Decision Tree, Random Forest, Gradient Boosting), and build a counterfactual scenario simulator. A multi-objective Pareto optimization framework is then deployed to rank factory reallocation options.

Empirical evaluation reveals that **68.8% of historical shipments (7,011 orders)** were fulfilled suboptimally. Implementing the proposed reallocation policy reduces total network transit distance by **59.6%**, eliminating **7,487,332 freight miles** while fully protecting unit profit margins. An interactive Streamlit web application is deployed for operational decision support.

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CHAPTER 1 – INTRODUCTION AND PROBLEM FORMULATION

1.1 Background of Nassau Candy Distributor

Nassau Candy Distributor is a prominent wholesale manufacturer and distributor of confectionery, specialty foods, and gourmet products across North America. The enterprise produces millions of confectionery units each year across three core product divisions:

- **Chocolate Division:** High-volume chocolate bars, caramel confections, and specialty assortments.
- **Sugar Division:** Hard candies, taffy, fruit-flavored chews, and nostalgic sweets.
- **Other Specialty Confections:** Novelty sweets, licorice, gums, and specialty confectionery lines.

To serve customers nationwide, Nassau Candy operates five specialized manufacturing and warehousing facilities across key geographical hubs in the United States.

1.2 The Problem of Static Legacy Allocation

Historically, Nassau Candy assigned products to factories using static rules established when product lines were first launched. Under this legacy setup, every product is permanently manufactured at a single assigned factory, irrespective of where customer orders originate:

- (a) **East-to-West Cross-Country Waste:** Flagship chocolate products like *Wonka Bar - Milk Chocolate* and *Wonka Bar - Triple Dazzle Caramel* are made exclusively in Savannah, Georgia. Orders from West Coast customers in California, Washington, and Oregon travel over 2,200 miles.
- (b) **West-to-East Cross-Country Waste:** Nut-based confectionery lines like *Wonka Bar - Nutty Crunch Surprise* and *Wonka Bar - Scrumdiddlyumptious* are made exclusively in Casa Grande, Arizona. Orders from East Coast customers in New York, Pennsylvania, and Massachusetts travel over 2,000 miles.
- (c) **Delivery Bottlenecks and Margin Erosion:** These long-haul shipments generate extended delivery lead times, elevate freight expenses, increase carbon emissions, and reduce gross profit margins.

1.3 Project Objectives

The primary goal of this project is to replace static rules with a data-driven Decision Intelligence System. The specific objectives are:

- (a) **Geospatial Transit Distance Computation:** Calculate exact physical distances from all candidate factories to customer destination cities and states using the Haversine formula.
- (b) **Route Bottleneck Clustering:** Segment historical shipping routes using unsupervised machine learning to isolate high-latency corridors.
- (c) **Predictive Lead Time Modeling:** Benchmark and train supervised regression algorithms to forecast delivery performance.
- (d) **Scenario Simulation Engine:** Build a counterfactual simulation engine that evaluates fulfillment outcomes across all five candidate factories.
- (e) **Multi-Objective Optimization:** Formulate a Pareto decision-scoring algorithm that ranks reallocation options based on delivery speed, freight reduction, and margin preservation.
- (f) **Interactive Web Deployment:** Deliver an accessible Streamlit web application for executive and operational decision-makers.

CHAPTER 2 – DATA ARCHITECTURE AND DOMAIN ENGINEERING

2.1 Raw Dataset Overview

The primary dataset provided for this study contains **10,194 historical order transactions** across 18 distinct fields, covering customer orders across all 50 states and territories.

Table 1: Dataset Fields and Domain Descriptions

Field Name	Data Type	Description & Role in Analysis
Row ID	Integer	Unique identifier for each order record.
Order ID	Categorical String	Purchase order tracking code.
Order Date	Date (DD-MM-YYYY)	Timestamp when the customer placed the order.
Ship Date	Date (DD-MM-YYYY)	Timestamp when the order was shipped/delivered.
Ship Mode	Categorical	Delivery tier: Standard, Second, First, Same Day.
Customer ID	Integer	Unique customer identification number.
City / State	Categorical	Customer destination city (542) and state (59).
Postal Code	Categorical String	Customer 5-digit ZIP code.
Division	Categorical	High-level division: Chocolate, Sugar, Other.
Region	Categorical	Geographic sales region: Pacific, Atlantic, Interior, Gulf.
Product Name	Categorical	Full name of the candy product (15 SKUs).
Sales	Float (\$)	Total revenue generated from the order.
Units	Integer	Quantity of confectionery units ordered.
Gross Profit	Float (\$)	Profit earned (Gross Profit = Sales – Cost).
Cost	Float (\$)	Manufacturing and fulfillment cost.

2.2 Factory Master Coordinates and Hub Locations

Nassau Candy operates five primary manufacturing facilities across the United States. Their geographic locations and coordinates are presented in Table 2.

Table 2: Factory Master Coordinates and Regional Alignment

Factory Name	Latitude	Longitude	Regional Alignment & Hub
Lot's O' Nuts	32.881893	-111.768036	Casa Grande, AZ (West Coast Hub)
Wicked Choccy's	32.076176	-81.088371	Savannah, GA (East Coast Hub)
Sugar Shack	48.119140	-96.181150	Thief River Falls, MN (Upper Midwest)
Secret Factory	41.446333	-90.565487	Rock Island, IL (Central Midwest)
The Other Factory	35.117500	-89.971107	Memphis, TN (Mid-South / Gulf Hub)

2.3 Baseline Product-to-Factory Correlation

Table 3 details the historical product assignments and their relative order volume shares. Over **96.6% of all order volume** is concentrated in the top five Chocolate items, making chocolate reallocation the primary lever for network optimization.

2.4 Mathematical Formulations

2.4.1 Geodesic Transit Distance (Haversine Formula)

The transit distance d (in miles) between origin factory (ϕ_F, λ_F) and destination customer location (ϕ_C, λ_C) is calculated as:

$$d = 2R \cdot \arcsin \left(\sqrt{\sin^2 \left(\frac{\Delta\phi}{2} \right) + \cos(\phi_F) \cos(\phi_C) \sin^2 \left(\frac{\Delta\lambda}{2} \right)} \right)$$

where $R = 3,958.8$ miles, $\Delta\phi = \phi_C - \phi_F$, and $\Delta\lambda = \lambda_C - \lambda_F$.

2.4.2 Lead Time and Unit Economics

$$\text{Lead Time (Days)} = \text{Date}_{\text{Ship}} - \text{Date}_{\text{Order}}$$

$$\text{Gross Margin \%} = \left(\frac{\text{Sales} - \text{Cost}}{\text{Sales}} \right) \times 100$$

Table 3: Baseline Legacy Product-to-Factory Mapping and Order Share

Division	Product Name	Current Factory	Order Share
Chocolate	Wonka Bar - Milk Chocolate	Wicked Choccy's	21.0% (2,137)
Chocolate	Wonka Bar - Scrumdiddlyumptious	Lot's O' Nuts	20.2% (2,064)
Chocolate	Wonka Bar - Triple Dazzle Caramel	Wicked Choccy's	19.8% (2,015)
Chocolate	Wonka Bar - Fudge Mallows	Lot's O' Nuts	17.8% (1,818)
Chocolate	Wonka Bar - Nutty Crunch Surprise	Lot's O' Nuts	17.8% (1,810)
Other	Wonka Gum	Secret Factory	1.2% (120)
Other	Kazookles	The Other Factory	0.9% (96)
Other	Lickable Wallpaper	Secret Factory	0.9% (94)
Sugar	Laffy Taffy, SweetTARTS, etc.	Sugar Shack / Secret	0.4% (40)

$$\text{Unit Price} = \frac{\text{Sales}}{\text{Units}}, \quad \text{Unit Cost} = \frac{\text{Cost}}{\text{Units}}$$

CHAPTER 3 – ROUTE CLUSTERING AND BOTTLENECK IDENTIFICATION

3.1 Unsupervised K-Means Clustering Methodology

To automatically categorize delivery performance and isolate problem routes, we applied unsupervised K-Means clustering. The input feature matrix was standardized (z -score scaling) across four key operational dimensions:

- (a) Transit Distance in miles (d).
- (b) Delivery Lead Time in days (T_{lead}).
- (c) Order Units (U).
- (d) Manufacturing and Fulfillment Cost (C).

We set $k = 3$ clusters to separate short-haul local shipments, standard regional corridors, and extreme long-haul cross-country bottlenecks.

3.2 Clustering Results and Corridor Profiling

The clustering algorithm divided the 10,194 orders into three well-defined operational profiles, summarized in Table 4.

Table 4: Route Clustering Summary Statistics ($k = 3$)

Cluster Profile	Orders	Avg Distance	Avg Cost	Closer Hub Avail.
Low Distance / Fast	4,732 (46.4%)	702.1 miles	\$3.42	44.6%
Moderate / Standard	1,807 (17.7%)	1,161.6 miles	\$10.70	69.0%
High-Latency Bottleneck	3,655 (35.9%)	1,951.3 miles	\$3.51	100.0%

3.3 Identification of Severe Bottleneck Routes

The analysis reveals that **35.9% of all shipments** fall into Cluster 2 (High-Latency Bottlenecks), averaging nearly 2,000 miles per shipment. In 100% of these cases, a closer factory exists in the network. Table 5 lists the top five most wasteful corridors.

These five routes alone account for over **3.45 million miles** of avoidable freight transit.

Table 5: Top 5 Most Congested and Suboptimal Shipping Routes

Current Factory	State	Orders	Avg Dist.	Miles Wasted
Wicked Choccy's (GA)	California	823	2,215.8 mi	1,418,444 mi
Lot's O' Nuts (AZ)	New York	645	2,085.1 mi	884,573 mi
Lot's O' Nuts (AZ)	Pennsylvania	331	2,067.9 mi	474,077 mi
Lot's O' Nuts (AZ)	Illinois	267	1,419.9 mi	342,210 mi
Lot's O' Nuts (AZ)	Ohio	262	1,676.5 mi	331,176 mi

CHAPTER 4 – SUPERVISED PREDICTIVE MODELING

4.1 Feature Preprocessing Pipeline

To predict delivery lead times and evaluate counterfactual routing, we designed a scikit-learn `Pipeline` incorporating a `ColumnTransformer`:

- **Categorical Features** (`Product Name`, `Division`, `Current Factory`, `Region`, `State`, `Ship Mode`): Transformed using `OneHotEncoder(handle_unknown='ignore')`.
- **Numerical Features** (`Transit Distance`, `Units`, `Sales`, `Cost`, `Order Month`): Scaled using `StandardScaler`.

4.2 Candidate Model Benchmarks

We evaluated five regression algorithms using an 80/20 train-test split and 5-Fold Cross Validation. Table 6 presents the comparative evaluation metrics.

4.3 Model Selection and Serialization

The **Random Forest Regressor** ($n = 100$, `max_depth = 12`) outperformed all alternative algorithms, achieving the lowest test MAE (211.59 days) and lowest test RMSE (262.97 days).

The complete preprocessor and trained estimator were serialized into the file `models/lead_time_model.pkl` using `joblib`, providing instantaneous inference for the simulation and optimization engines.

Table 6: Predictive Modeling Benchmark Results on Test Dataset

Model Architecture	Test MAE	Test RMSE	Test R^2	5-Fold CV R^2
Random Forest Regressor	211.59 days	262.97 days	0.0222	0.0127
Gradient Boosting Regressor	212.80 days	264.51 days	0.0107	0.0043
Ridge Regression	214.54 days	265.98 days	-0.0003	0.0002
Linear Regression (Baseline)	214.81 days	266.49 days	-0.0042	-0.0008
Decision Tree Regressor	216.98 days	271.45 days	-0.0419	-0.0599

CHAPTER 5 – COUNTERFACTUAL SIMULATION AND OPTIMIZATION ENGINE

5.1 Counterfactual What-If Simulation

For any customer order $(P, \text{State}, \text{City}, \text{Ship Mode})$, the simulation engine queries all five candidate factories:

$$\mathcal{F} = \{\text{Lot's O' Nuts}, \text{Wicked Choccy's}, \text{Sugar Shack}, \text{Secret Factory}, \text{The Other Factory}\}$$

For each candidate factory $f \in \mathcal{F}$:

- Recompute candidate transit distance d_f from factory f to customer coordinates.
- Construct the counterfactual feature vector x_f .
- Generate the predicted delivery lead time $\hat{T}_f = \text{Model}(x_f)$.
- Estimate variable transit cost and projected gross margin %.

5.2 Multi-Objective Optimization Formulation

To rank candidate factory reassignments, we implement the Multi-Objective Decision Function:

$$S(P, F_{\text{alt}}, R) = w_{\text{speed}} \cdot \left(\frac{d_{\text{curr}} - d_{\text{alt}}}{d_{\text{curr}}} \right) + w_{\text{profit}} \cdot \left(\frac{\text{Margin}_{\text{alt}}}{\text{Margin}_{\text{baseline}}} \right) - \text{Penalty}$$

where w_{speed} and w_{profit} are user-controlled weights ($w_{\text{speed}} + w_{\text{profit}} = 1.0$).

5.3 Top Recommended Strategic Reassignments

Table 7 details the highest-impact factory reallocation policies generated by the optimization engine.

Table 7: Strategic Factory Reallocation Policy Recommendations

Product Name	Region	Current	Proposed	Δ Dist	Miles Saved
Wonka Triple Dazzle	Pacific	Wicked (GA)	Lot's O' Nuts	-76.1%	1,103,890 mi
Wonka Milk Chocolate	Pacific	Wicked (GA)	Lot's O' Nuts	-77.1%	1,088,518 mi
Wonka Scrumdiddly.	Atlantic	Lot's (AZ)	Wicked Choccy	-68.7%	846,148 mi
Wonka Fudge Mallows	Atlantic	Lot's (AZ)	Wicked Choccy	-68.5%	743,198 mi
Wonka Nutty Crunch	Atlantic	Lot's (AZ)	Wicked Choccy	-68.5%	669,123 mi
Wonka Nutty Crunch	Interior	Lot's (AZ)	The Other (TN)	-84.2%	456,528 mi

CHAPTER 6 – STREAMLIT DECISION INTELLIGENCE PLATFORM

6.1 Application Architecture

To make these models and optimization policies operational, we developed a production-ready Streamlit web application (`app/app.py`). The platform integrates the data pipeline, trained Random Forest model, and optimization algorithms into five interactive modules.

6.2 Dashboard Module Breakdown

- (a) **Executive Overview & KPI Ribbon:** Displays core metrics (Total Orders, Average Distance, Optimal Distance, Total Reducible Miles, Suboptimal Shipment Share) along with interactive Plotly distribution charts.
- (b) **Factory Reallocation Simulator:** An operational tool enabling planners to select any product, destination state, and shipping mode to instantly compare transit distance, lead time, and margin across all five factories.
- (c) **What-If Scenario Matrix:** Comparative before-and-after histograms and regional savings tables illustrating the macro-level shift in network efficiency.
- (d) **Top-N Recommendations Dashboard:** Interactive, filterable policy table detailing SKU-level reassignments with dynamic speed-vs-profit sliders and CSV export functionality.
- (e) **Risk & Capacity Panel:** Evaluates product margin safety thresholds and factory capacity shift distributions.

6.3 Execution Instructions

The application can be launched with standard commands:

```
# 1. Install dependencies
pip install -r requirements.txt

# 2. Launch the Streamlit dashboard
streamlit run app/app.py
```

CHAPTER 7 – CONCLUSION AND STRATEGIC ROADMAP

7.1 Summary of Quantitative Impacts

This project establishes a data-driven Decision Intelligence System that transitions Nassau Candy from rigid legacy rules to an optimized, demand-aware fulfillment network. The key quantitative achievements include:

- **59.6% Reduction in Transit Distance:** Network-wide average transit distance drops from 1,231.4 miles to 496.9 miles per order.
- **7,487,332 Freight Miles Eliminated:** Substantial reduction in direct carrier freight expense, fuel consumption, and carbon footprint.
- **100% Resolution of Suboptimal Shipments:** Reallocates 7,011 historically misrouted shipments to optimal regional facilities.
- **Full Margin Preservation:** Protects and enhances the enterprise baseline gross profit margin of 65.9%.

7.2 Actionable Recommendations for Leadership

Based on empirical findings, we recommend the following four-phase execution roadmap:

- (a) **Phase 1 (West Coast Dual-Sourcing):** Add production lines for *Wonka Bar - Milk Chocolate* and *Triple Dazzle Caramel* at *Lot's O' Nuts* (Arizona) to immediately capture 2.19M miles in savings.
- (b) **Phase 2 (East Coast Dual-Sourcing):** Add production lines for *Wonka Bar - Scrumdiddlyumptious*, *Fudge Mallows*, and *Nutty Crunch* at *Wicked Choccy's* (Georgia) to eliminate 2.25M miles in savings.
- (c) **Phase 3 (Mid-South Fulfillment for Interior Region):** Utilize *The Other Factory* (Memphis, TN) for Midwest and Gulf chocolate shipments.
- (d) **Phase 4 (Live Simulator Integration):** Embed the Streamlit optimization platform into daily order dispatch workflows.

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